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CHAPTER 03
General Vector Spaces

Part 2

Span of a Set of Vectors

- Let V be a vector space and let $S = \{\mathbf{v}_1, \ldots \mathbf{v}_n\}$ be a (finite) set of vectors in V.
- The **span** of S, denoted by **span**(S) or span $\{\mathbf{v}_1, \ldots \mathbf{v}_n\}$, is the set:

$$\textbf{span}(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots\ldots\ldots\ldots\ldots + c_n\mathbf{v}_n \mid c_1, c_2, \ldots\ldots\ldots, c_n \in \Re\}$$

- In other word, determine if every vector in V can be expressed as **linear combination** of the vectors in S or not.
- span(S) is a subspace of V.

The span of single vector

- The span of a single nonzero vector in $\Re^n$ is a **line** through the origin as shown in figure 1(a).

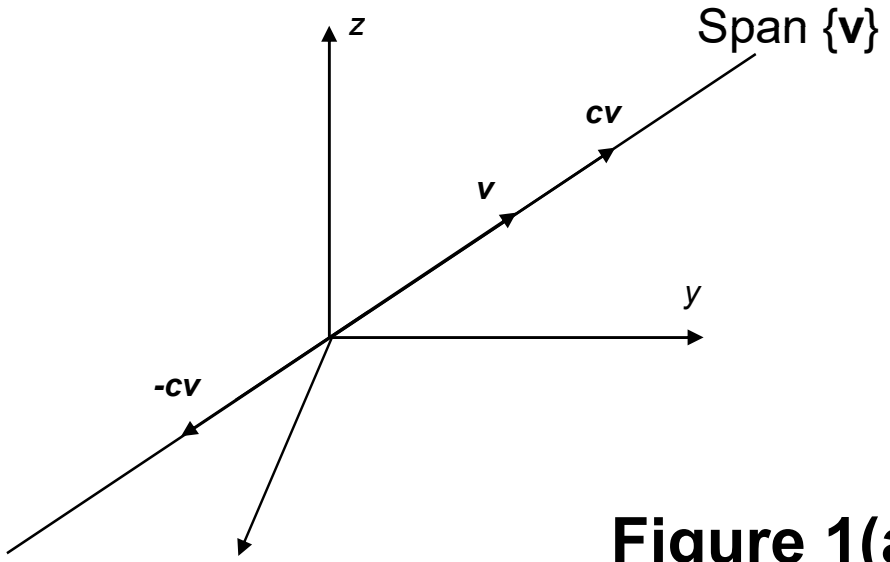


Figure 1(a)

The span of two vectors

- The span of two linearly independent vectors is a **plane** through the origin as shown in figure 1(b).

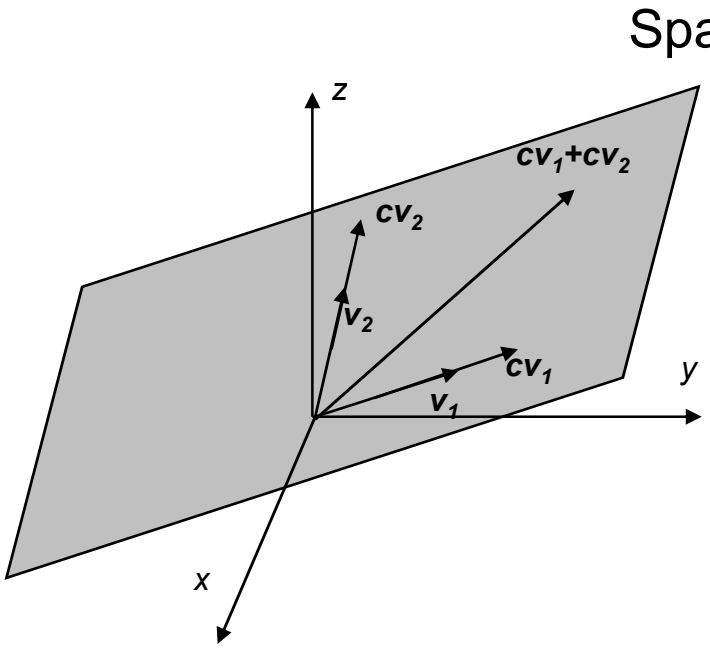


Figure 1(b)

Example #1

- Let S be the subset of the vector space $\Re^3$ defined by

$$S = \left\{ \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} \right\}$$

Show that $v = \begin{bmatrix} -4 \\ 4 \\ -6 \end{bmatrix}$

is in **span**(S).

Solution

- The vector, **v** is in **span**(*S*) provided that there are scalars c_1, c_2 **and** c_3 such that

$$c_1\begin{bmatrix}2\\-1\\0\end{bmatrix}+c_2\begin{bmatrix}1\\3\\-2\end{bmatrix}+c_3\begin{bmatrix}1\\1\\4\end{bmatrix}=\begin{bmatrix}-4\\4\\-6\end{bmatrix}$$

- This linear system in matrix form is given by

$$\left[\begin{array}{ccc|c}2 & 1 & 1 & -4\\-1 & 3 & 1 & 4\\0 & -2 & 4 & -6\end{array}\right]$$

Solution (cont.)

- After row-reducing, we obtain

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right] \Rightarrow \begin{aligned} c_1 &= -2 \\ c_2 &= 1 \\ c_3 &= -1 \end{aligned}$$

Since every vector in $\Re^3$ can be written as a linear combination of the three given vectors, thus the vector **v** is in **span(*S*)**.

Solution

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & -4 \\ -1 & 3 & 1 & 4 \\ 0 & -2 & 4 & -6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -3 & -1 & -4 \\ 2 & 1 & 1 & -4 \\ 0 & -2 & 4 & -6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -3 & -1 & -4 \\ 2 & 1 & 1 & -4 \\ 0 & -2 & 4 & -6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -3 & -1 & -4 \\ 0 & 7 & 3 & 4 \\ 0 & -2 & 4 & -6 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -3 & -1 & -4 \\ 0 & 1 & 3/7 & 4/7 \\ 0 & -2 & 4 & -6 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 2/7 & -16/7 \\ 0 & 1 & 3/7 & 4/7 \\ 0 & 0 & 34/7 & -34/7 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & -1/7 & -16/7 \\ 0 & 1 & 3/7 & 4/7 \\ 0 & 0 & 1 & -1 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$c_1 = -2$, $c_2 = 1$ and $c_3 = -1$

This shows that v is a linear combination of the vectors in S and is thus in **span**(*S*)

Example #2

Show that

$$\textit{span}\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix},\begin{bmatrix}-2\\3\\1\end{bmatrix},\begin{bmatrix}1\\2\\4\end{bmatrix}\right\}=\boldsymbol{R}^3$$

Solution

Let $\mathbf{v} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an arbitrary element of $\Re^3$.

The vector v is in **span**(*S*) provided that there are scalars c_1, c_2 **and** c_3 such that

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Solution (cont.)

$$\left[\begin{array}{ccc|c} 1 & -2 & 1 & x_1 \\ -1 & 3 & 2 & x_2 \\ 0 & 1 & 4 & x_3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 1 & x_1 \\ -1 & 3 & 2 & x_2 \\ 0 & 1 & 4 & x_3 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -2 & 1 & x_1 \\ 0 & 1 & 3 & x_1 + x_2 \\ 0 & 0 & 1 & x_3 - x_1 - x_2 \end{array}\right]$$

This shows that a solution exists for all choice of x_1, x_2 and x_3 . Thus every vector in $\Re^3$ can be written as a linear combination of the three given vectors. Hence, the span of the three vectors is all of $\Re^3$.

Exercise

1) Determine whether $\boldsymbol{v}_1=(-1,2,1)$, $\boldsymbol{v}_2=(4,1,-3)$ and $\boldsymbol{v}_3=(-6,3,5)$ span the vector space $\Re^3$.

2) Show that

$$\mathbf{span}\left\{ \begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\0\\2\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix} \right\} = \Re^3$$

Basis for a Vector Space

- A subset ***B*** of a vector space *V* is a basis for *V* provided that
  1) ***B*** is linearly independent set of vectors in *V*
  2) **span**(***B***) = *V*

- For any $\Re^n$,

$$\varepsilon_n = \left\{ \begin{bmatrix} 1 \\ 0 \\ . \\ . \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ . \\ . \\ 0 \end{bmatrix}, \ldots\ldots\ldots \ldots \begin{bmatrix} 0 \\ 0 \\ . \\ . \\ 1 \end{bmatrix} \right\}$$

is the standard basis and denote by $\mathbf{e}_1, \mathbf{e}_2, \ldots\ldots \mathbf{e}_n$

Example #3

- Show that the set

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

is the basis for $\mathfrak{R}^3$.

Solution

1) Show that **span**(B) = V

$$c_1\begin{bmatrix}1\\1\\0\end{bmatrix}+c_2\begin{bmatrix}1\\1\\1\end{bmatrix}+c_3\begin{bmatrix}0\\1\\-1\end{bmatrix}=\begin{bmatrix}a\\b\\c\end{bmatrix}\Rightarrow\left[\begin{array}{ccc|c}1&1&0&a\\1&1&1&b\\0&1&-1&c\end{array}\right]=\left[\begin{array}{ccc|c}1&0&0&2a-b-c\\0&1&0&-a+b+c\\0&0&1&-a+b\end{array}\right]$$

This show that a solution exists for all choices of a, b and c. Thus, **span**(B) = $\Re^3$

Solution (cont.)

2) Show that *B* is linearly independent set of vectors in *V*

$$c_1\begin{bmatrix}1\\1\\0\end{bmatrix}+c_2\begin{bmatrix}1\\1\\1\end{bmatrix}+c_3\begin{bmatrix}0\\1\\-1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}\Rightarrow\left[\begin{array}{ccc|c}1&1&0&0\\1&1&1&0\\0&1&-1&0\end{array}\right]=\left[\begin{array}{ccc|c}1&0&0&0\\0&1&0&0\\0&0&1&0\end{array}\right]$$

This implies that c_1 =0, c_2 =0 and c_3 = 0.

Thus set *B* is linearly independent.

Solution (cont.)

- Alternatively, compute the determinant of the coefficients matrix and set of vectors is linearly independent if and only if the determinant is nonzero.

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = 1$$

Hence the set *B* is a basis for $\Re^3$

Example # 4

Determine whether

$$B = \left\{ \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -4 \end{bmatrix} \right\}$$

is the basis for M_{2x2}

Solution

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, Show that **A** is in **span**(*B*)

$$c_1\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} + c_2\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} + c_3\begin{bmatrix} 0 & 1 \\ 0 & -4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The augmented matrix is $\left[\begin{array}{ccc|c} 1 & -1 & 0 & a \\ 3 & 2 & 1 & b \\ 2 & 1 & 0 & c \\ 1 & 0 & -4 & d \end{array}\right]$

Solution(cont)

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & a \\ 3 & 2 & 1 & b \\ 2 & 1 & 0 & c \\ 1 & 0 & -4 & d \end{array}\right] = \left[\begin{array}{ccc|c} 1 & -1 & 0 & a \\ 0 & 5 & 1 & -3a+b \\ 0 & 0 & -3 & -a-3b+5c \\ 0 & 0 & 0 & a+4b-7c+d \end{array}\right]$$

- Observe that the linear system is consistent only if $a+4b-7c+d = 0$. Here, the linear system in inconsistent.

Hence, B **does not** span M_{2x2} . **Therefore, the set *B* is not a basis for M_{2x2}**

Dimension

- The dimension of a finite dimensional vector space, denoted by *dim(V),* is defined to be the number of vectors in a basis for *V*.

- In additional, the *zero vector space* is defined to have dimensional *zero*.

Dimension

- $\dim(\Re^n) = n$

The standard basis has n vectors $(e_1, e_2, \ldots e_n)$.

- $\dim(M_{2x2}) = 4$

The standard basis of $M_{2\times2}$ has 4 vectors.

- $\dim(M_{mxn}) = m*n$

The standard basis has $m*n$ vectors.

- $\dim(P_n) = n+1$

The standard basis has $n+1$ vectors.

Example

Determine a basis and the dimension of the solution space of the homogeneous system.

$$2x_1 + 2x_2 - 3x_3 + 0x_4 + x_5 = 0$$
$$-x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0$$
$$x_1 + x_2 - 2x_3 + 0x_4 - x_5 = 0$$
$$0x_1 + 0x_2 + x_3 + x_4 + x_5 = 0$$

Solution

Find the solution space of the homogeneous system by using Gaussian elimination method.

$$\left(\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right) \Rightarrow \left(\begin{array}{ccccc|c} 2 & 2 & -3 & 0 & 1 & 0 \\ 0 & 0 & 0.5 & -3 & 1.5 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 \end{array}\right).$$

It shows that x_1, x_3, x_4 and x_5 are leading variables in which $x_3=x_4=x_5=0$ and x_2 is a free variable. Let $x_2= a$, then we obtain $x_1=-x_2=-a$.

Solution (cont.)

Write the solution as follows.

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -a \\ a \\ 0 \\ 0 \\ 0 \end{pmatrix} = a \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow A = \left\{ x \mid x = a \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, a \in R \right\}$$

Solution (cont.)

This is an evident that the vector **v** is spans the solution space *A* and linearly independent which implies that **v** is a basis of *A.*

$$\mathbf{v} = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Therefore, the dimension of the solution space *A* is equal to 1 since *A* contains only one vector basis.

Exercise

1) Determine the dimension of and a basis for the solution space of the system

$$
\begin{aligned}
2x_1 + 2x_3 - x_3 + x_5 &= 0 \\
-x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\
x_1 + x_2 - 2x_3 - x_5 &= 0 \\
x_3 + x_4 + x_5 &= 0
\end{aligned}
$$

2)Find the basis for the subspace *S* of the vector space *V* and specify the dimension of *S*

$$
S = \left\{ \begin{pmatrix} s+2t \\ -s+t \\ t \end{pmatrix} \middle| \; s,t \in \Re \right\} V = \Re^3
$$

End Part 2